# ClipSAT : Contest Math & Olympiad AMC 10/12, Kangaroo & Mawhiba

## Unit 8.01: Invariants, Monovariants, Coloring & Combinatorial Games Name _____

**Q1.** The coordinate diagram shows the parabolic curve f_1x = x^2 - 20x + 118 in the domain of Invariants, Monovariants, Coloring & Games. By completing the square f_1x = x - h^2 + k , find the sum k + f_112 .

[ ] A: 50
[ ] B: 22
[ ] C: 38
[ ] D: 40

**Q2.** A mathematical model in Invariants, Monovariants, Coloring & Games is defined by the cubic function y_2x = 3x^3 - 10x^2 + 15x - 15 . Calculate the exact gradient of the tangent to the curve at the point where x = 3 .

[ ] A: 36
[ ] B: 96
[ ] C: 21
[ ] D: 22

**Q3.** In Invariants, Monovariants, Coloring & Games, two displacement vectors are given by mathbf{u}_{3} = langle 10, -8 rangle and mathbf{v}_{3} = langle 9, 7 rangle . Compute the exact scalar value of mathbf{u}_{3} cdot mathbf{v}_{3} + |mathbf{u}_{3}|^2 .

[ ] A: 34
[ ] B: 198
[ ] C: 164
[ ] D: 310

**Q4.** The linear growth profile shown in the diagram for Invariants, Monovariants, Coloring & Games follows an arithmetic progression with first term a_1 = 11 and common difference d = 6 . Find the exact sum S_{15} of the first 15 terms.

[ ] A: 95
[ ] B: 885
[ ] C: 795
[ ] D: 1425

**Q5.** In Invariants, Monovariants, Coloring & Games, evaluate the exact definite integral I_5 = int_0^{2} 18x^2 + 40x,mathrm{d}x .

[ ] A: 152
[ ] B: 168
[ ] C: 65
[ ] D: 128

**Q6.** In Invariants, Monovariants, Coloring & Games, consider the quadratic function f_6x = x^2 - 22x + 211 . Express f_6x in the completed-square form x - h^2 + k and compute the exact value of k + f_618 .

[ ] A: 229
[ ] B: 240
[ ] C: 139
[ ] D: 222

**Q7.** The graph illustrates the cubic model y_7x = 3x^3 - 11x^2 + 33x - 22 studied in Invariants, Monovariants, Coloring & Games. Find the instantaneous gradient frac{mathrm{d}y}{mathrm{d}x} of the tangent to the curve at x = 8 .

[ ] A: 609

[ ] B: 433

[ ] C: 400

[ ] D: 83

**Q8.** In Invariants, Monovariants, Coloring & Games, two displacement vectors are given by mathbf{u}_{8} = langle 11, -7 rangle and mathbf{v}_{8} = langle 8, 5 rangle . Compute the exact scalar value of mathbf{u}_{8} cdot mathbf{v}_{8} + |mathbf{u}_{8}|^2 .

[ ] A: 53

[ ] B: 170

[ ] C: 223

[ ] D: 293